

Degree Program: Electrical and Computer Engineering Course Code: CSE425 Course Title: Concepts of Programming Language		
Credit: 3.0 (Theory)	Terms Offered: Summer	
Exam Hours: 3.00	CIE Marks: 30%	SEE Marks: 70%

Course Objectives / Summary: This course covers the fundamental concepts, principles, and paradigms underlying modern programming languages. Students will explore the syntax and semantics of programming languages, data types, control structures, subprograms, and abstraction mechanisms. The course examines multiple programming paradigms including imperative, object-oriented, functional, and logic programming, enabling students to understand the design rationale and implementation trade-offs of different language features. Through comparative analysis of programming languages, students will develop the ability to evaluate language features, select appropriate languages for specific problems, and apply programming paradigms effectively. Besides, studying programming language concepts enhances critical thinking about software design, prepares students to learn new languages quickly, and provides a foundation for compiler design and software engineering practices.

Course Learning Outcomes (CLOs): at the end of the Course, the student will be able to-

CLO 1	Explain the fundamental concepts of programming languages including syntax and semantics, names, bindings, scopes, data types, expressions, control structures, and subprograms; describe different programming paradigms (imperative, object-oriented, functional, logic) and their characteristics.
CLO 2	Apply programming language concepts to write and interpret code in different paradigms; utilize parameter passing mechanisms, scoping rules, type systems, and control structures appropriately in program design.

CLO 3	Analyze and compare different programming language features, design choices, and paradigms; evaluate language suitability for specific problem domains based on readability, writability, reliability, and performance criteria.
CLO 4	Design solutions using appropriate programming paradigms and language features; construct programs incorporating abstract data types, object-oriented principles, functional programming techniques, and logic programming concepts for solving complex problems.

**** CLO maintains the following distinct cognitive levels:**

CLO 1: Lower-order (Explain/Discuss) - Understanding

CLO 2: Mid-order (Apply/Utilize) - Application

CLO 3: Higher-order (Analyze/Evaluate) - Analysis

CLO 4: Highest-order (Construct/Design/Formulate) - Synthesis/Creation

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes (PLOs)-

Course Learning Outcomes (CLOs)	Program Learning Outcomes (PLOs)											
	01	02	03	04	05	06	07	08	09	10	11	12
CLO1	3	2										2
CLO2	3	2			3							
CLO3		3		2	3							
CLO4	2		3		3							

(Tick mark or level of correlation: 3-High, 2-Medium, 1-Low can be used)

SN	COURSE CONTENT	Hrs.	CLOs
1	Introduction to Programming Languages: Historical development of programming languages, Language evaluation criteria (readability, writability, reliability, cost), Language design trade-offs, Implementation methods (compilation, interpretation, hybrid), Programming paradigms overview (imperative, object-oriented, functional, logic).	03	CLO1
2	Syntax and Semantics: Lexical structure and tokens, Context-free grammars and BNF notation, Parse trees and abstract syntax trees, Attribute grammars, Operational semantics, Denotational semantics, Axiomatic semantics.	04	CLO1, CLO3
3	Names, Bindings, and Scopes: Variables and constants, Naming conventions, Binding time and storage bindings, Type binding and type inference, Scope (static vs. dynamic scoping), Lifetime and garbage collection, Referencing environments.	03	CLO1, CLO2
4	Data Types: Primitive data types, Structured data types (arrays, records, strings), Pointers and references, Type checking and type compatibility, Type conversion and casting, User-defined types, Abstract data types.	03	CLO1, CLO2
5	Expressions and Assignment Statements: Arithmetic expressions, Operator precedence and associativity, Type conversions in expressions, Relational and Boolean expressions, Short-circuit evaluation, Assignment statements and multiple assignments.	02	CLO2
6	Control Structures: Statement-level control structures, Selection statements (if, switch/case), Iterative statements (for, while, do-while), Unconditional branching and goto, Guarded commands, Exception handling and assertions.	03	CLO1, CLO2

SN	COURSE CONTENT	Hrs.	CLOs
7	Subprograms: Fundamentals of subprograms, Parameters and parameter passing (pass-by-value, pass-by-reference, pass-by-result, pass-by-name), Local referencing environments, Overloaded subprograms, Generic subprograms, Closures, Coroutines.	04	CLO2, CLO3
8	Abstract Data Types and Encapsulation: Abstraction concepts, Encapsulation and information hiding, Abstract data type design issues, Language support for ADTs, Parameterized ADTs, Naming encapsulations (modules, packages, namespaces).	03	CLO3, CLO4
9	Object-Oriented Programming: Object-oriented concepts, Inheritance (single, multiple, interfaces), Dynamic binding and polymorphism, Encapsulation in OOP, Support for OOP in languages (Java, C++, Python), Design issues in OOP.	04	CLO2, CLO3, CLO4
10	Functional Programming: Introduction to functional programming, Mathematical functions and lambda calculus, Functional forms and higher-order functions, Recursion, Lists and list processing, Functional programming in languages (Scheme, ML, Haskell basics), Applications of functional programming.	04	CLO1, CLO3, CLO4
11	Logic Programming: Introduction to logic and logic programming, Predicate calculus and Horn clauses, Resolution and unification, Prolog basics (facts, rules, queries), List processing in Prolog, Backtracking, Applications of logic programming.	04	CLO1, CLO3, CLO4
12	Concurrency and Parallel Programming: Introduction to concurrency, Synchronization and race conditions, Semaphores and monitors, Message passing, Threads and parallel execution, Language support for concurrency.	02	CLO1, CLO3

Teaching Learning Strategies: Classroom lecture (white board and power point presentation), reading, solving practical problems and feedback.

Textbooks

1. *Concepts of Programming Languages*, Robert W. Sebesta, Pearson (11th or 12th Edition).
2. *Programming Language Pragmatics*, Michael L. Scott, Morgan Kaufmann/Elsevier (4th Edition).

Reference Books

1. *Programming Languages: Principles and Paradigms*, Maurizio Gabbriellini and Simone Martini, Springer (2nd Edition).
2. *Essentials of Programming Languages*, Daniel P. Friedman and Mitchell Wand, MIT Press (3rd Edition).

Assessment Pattern

CIE- Continuous Internal Evaluation (Marks: 90)

Bloom's Category	Class Tests (60 Marks) (%)	In class Participation (30 Marks) (%)
Remember	10	50
Understand	20	50
Apply	20	
Analyze	30	
Evaluate	20	
Create		

SEE- Semester End Examination (Marks: 210)

Bloom's Category	Final Exam (Marks 210) (%)
Remember	10
Understand	20
Apply	25
Analyze	20
Evaluate	20
Create	05